

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I T. Sai Varshini/192072403 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Sai Varshini

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Annexure A – Application Based Questions

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.

Identify email addresses.

Extract mobile numbers.

Detect technical skills (Python, Java, SQL, Machine Learning, NLP).

Extract years of experience.

Generate a structured summary of the candidate's profile.

Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.

Perform prefix-based searches.

Perform suffix-based searches.

Search using partial keywords.

Perform case-insensitive searches.

Display all matching product names.

Generate a report showing the total number of matching products for each search.

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.

Validate the institutional email address.

Validate the course code.

Validate the semester information.

Validate the student's mobile number.

Display appropriate validation messages for each field.

Generate a final registration status report indicating whether the registration is successful.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO-1 Assessment-1

Name :- T-Sai Varshini

Faculty: Dr. V. K. Gnanarej

Reg no :- 192472403

Course code/Name :- DSA0303 - Natural language processing

1) Resume Information Extraction System

1. Problem Context

Nowadays, Companies receive many resume for job recruitment. checking every resume manually takes a lot of time and effort. An automated Resume Information Extraction system helps the HR department by extraction system helps important details such as the candidates's name, email, mobile number etc. It also checks whether the candidate is eligible based on the required criteria.

2. Objectives

To develop a python program using Regular Expressions (Regex) that extracts candidate information from a resume and display only eligible candidate who have python skill and at least 2 years of experience.

3. Concepts used (Application)

- Natural language processing (NLP): used to process text data
- Regular Expressions (Regex): used to search and extract specific information from the resume.
- Python re Module: Provides functions like `re.search()` and `re.findall()` for pattern matching

4. Algorithm

1. Import the re module

2. Read the resume text
3. Extract the name, email, mobile number, skills, and experience using Regex.
4. Display the extracted details
5. Check if the candidate has python skill and at least 2 years of experience
6. Display whether the candidate is Eligible or Not Eligible.

5. Python program

```
import re
resume = " "
Name : Sai Varshini
Email : saivarshini@gmail.com
Mobile : 9876543210
Skills : python, SQL, NLP
Experience : 3 years
" " "
name = re.search(r"name: |s*(.+) ", resume)
email = re.search(r"[lw].-]+@[lw].-]+[lw]+ ", resume)
mobile = re.search(r"\d{10}", resume)
skills = re.findall(r"python|Java|SQL|machine learning|NLP", resume)
experience = re.search(r"(\d+)|s+years", resume)
print("Candidate Name : ", name.group(1))
print("Email : ", email.group(1))
print("Mobile : ", mobile.group(1))
print("Skills : ", " ".join(skills))
print("Experience : ", experience.group(1), "years")
if "python" in skills and int(experience.group(1)) >= 2 :
    print("Eligibility : Eligible")
else :
    print("Eligibility : NOT Eligible")
```

6. Sample Test case

Input: Name: Sai Varshini

Email: SaiVarshini@gmail.com

Mobile: 9876543210

Skills: Python, SQL, NLP

Experience: 3 years.

7. output: Candidate Name: Sai Varshini

Email: SaiVarshini@gmail.com

Mobile: 9876543210

Skills: Python, SQL, NLP

Experience: 3 years

Eligibility: Eligible.

Result :- The Python program successfully extracts the candidates details.

2) Product Search System

1. Problem Context

An e-commerce company offers a large number of products to customers. Finding a product using only an exact keyword can be difficult. Therefore, a product search system is required to support different search methods such as exact, prefix, suffix, partial, and case-insensitive searches.

This improves the user experience and helps customers find products quickly.

2. objectives

To develop a Python program using Regular Expression (Regex) that searches product names using different search techniques and displays all matching products along with the total number of matches.

3. Concepts Used (Application)

- Regular Expressions (Regex): used to search product names using different patterns.
- Pattern matching: Matches exact, prefix, suffix, and partial keywords.
- Python re module: uses functions like `re.search()` and `re.IGNORECASE` for efficient searching.

4. Algorithm

1. Import the re module
2. Stores product names in a list
3. Accept a search keywords from the user
4. Search the products using Regular Expressions.
5. Display all matching product names.
6. Display the total number of matching products.

5. Python Program

```
import re
Products = ["Laptop", "Laptop Bag", "Wireless mouse", "Bluetooth Speaker", "Smart watch", "USB Cable"]
Keyword = input("Enter product keyword: ")
matches = []
for product in products:
```

```
if re.search(keyword, product, re.IGNORECASE):  
    matches.append(product)  
print("\n Matching products: ")  
for item in matches:  
    print(item)  
print("\n Total matches: ", len(matches))
```

6. Sample Test Case

Input: Enter product keyword: lap

output: Matching products:

laptop

laptop Bag

Total matches: 2

7. Result

The product search system successfully searches product names using Regular Expressions and displays all matching products with the total number of matches. The system provides a simple and flexible product searching solution.

3) University Registration Validation System

1. Problem

Universities require students to enter correct information during online registration. Invalid register numbers, email addresses, course code, or mobile numbers can lead to registration errors.

Therefore, a validation system is required to verify student information before registration is completed.

2. Objective

To develop a Python program using Regular Expression (Regex)

3. Concepts used (Application)

- Regular Expressions (Regex): Used to validate different input formats
- Pattern Matching: Ensures the entered information follows the required format.
- Python re module: uses `re.match()` to validate user inputs

4. Algorithm

1. Import the re module
2. Read student details
3. Validate register number
4. Validate institutional email
5. Validate course code.
6. Validate semester
7. Validate mobile number
8. Display validation message
9. Display the final registration status

5. Python program

```
import re
```

```
reg = input("Register No: ")
```

```
email = input("Email: ")
```

```
mobile = input("Mobile: ")
```

```
if re.match(r"\d{2,3}[A-Z]{3,3}\d{4,9}", reg):
```

```
    Print("Register Number: valid")
```

else:

print("Register Number: Invalid")

if re.match(r".+@sarcetha1.com", email):

print("Email: Valid")

else:

print("Email: Invalid")

if re.match(r"\d{10}", mobile):

print("Mobile: Valid")

else:

print("Mobile: Invalid")

6. Sample Input : Register no : 23CSE1001
Email : student@sarceth.com
Mobile : 9876543210

Output : Register no : Valid
Email : Valid
Mobile : Valid.

7. Result : The program successfully validates all.